



Forecasting Solar to Plan the Site Battery

DKASC Alice Springs · Assessment 3 Technical Demonstration

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2 Overview: the problem we are solving

Abhisek



The site

DKASC Alice Springs: a solar array and, since 2024, a site battery (354 kWh, measured from its own data).



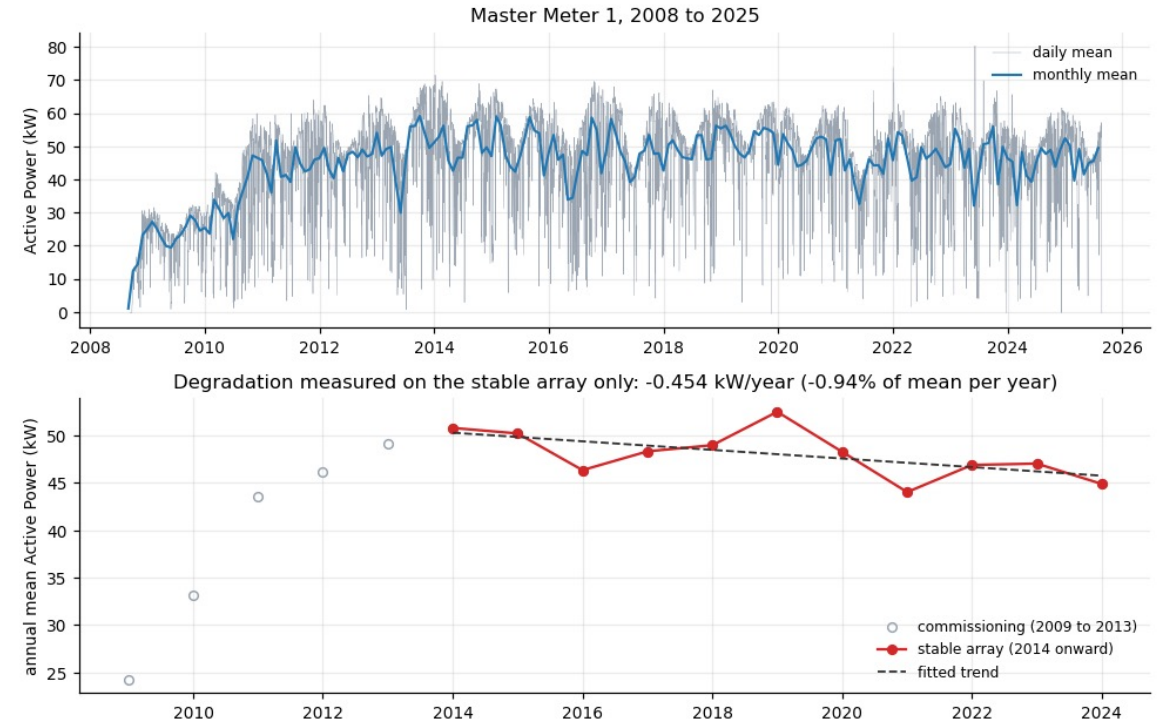
The data

Master Meter 1 data every 5 minutes from 2008 to 2025 (1.77 million rows) and site telemetry data for 2024 and 2025 (battery, solar, demand, and grid).



The decision

Each night: how much cheap off peak grid energy should the battery store for tomorrow's expensive evening peak?



Generation history at the site (A2, Figure 1)

Delivered in A2

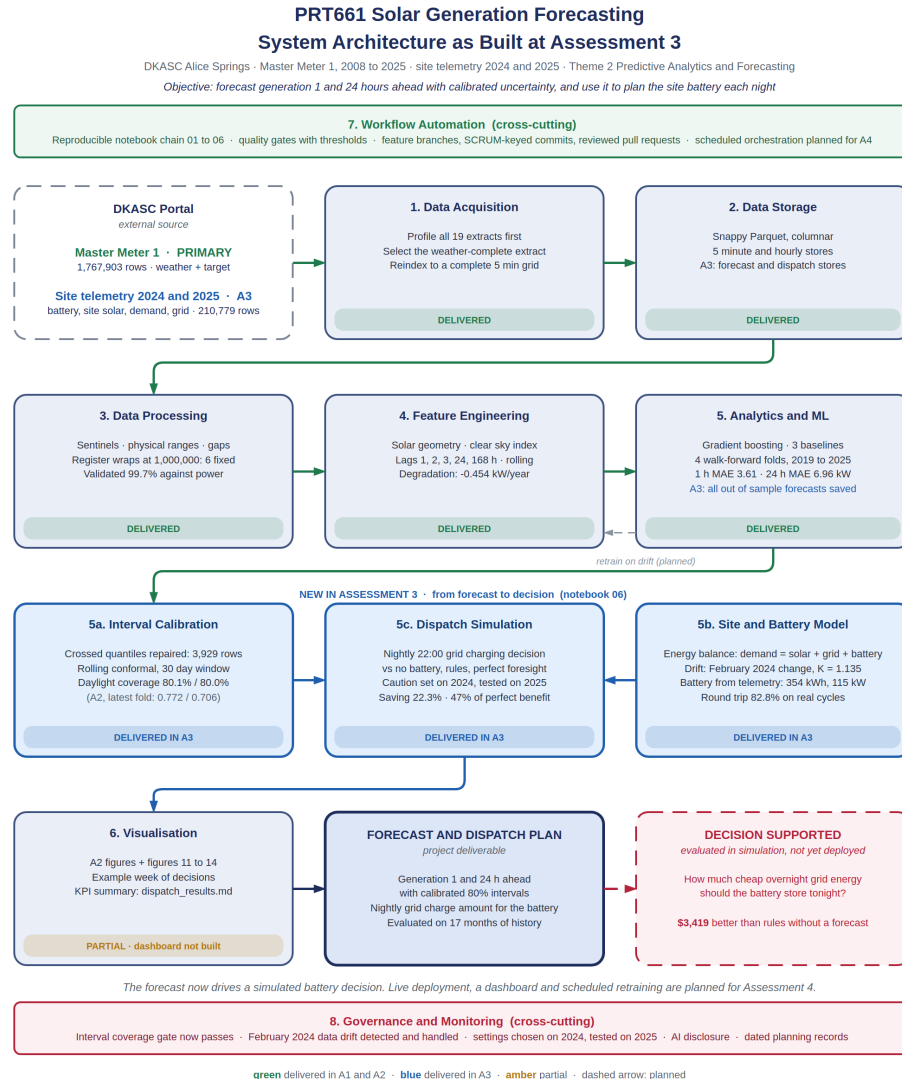
- Data pipeline from 19 raw extracts to a 30 feature hourly store
- Forecasts of 1 h and 24 h ahead using gradient boosting outperform three baselines
- 80% prediction intervals, but at 24 h they only covered 70.6% of hours in the latest test period.
- The actual battery decision was not yet modelled.



A3 objectives

- 1 Make the uncertainty credible: an 80% range that actually encompasses 80% of the values.
- 2 Model real site and battery from telemetry
- 3 Make the forecast a nightly battery decision and quantify the value.
- 4 Deliver it through Jira tasks, branches and reviewed pull requests

Updated system architecture



Two data sources

Master meter 1 for forecast, site telemetry 2024 and 2025 for the battery.

Stages 1 to 5 from A1 and A2

acquisition, storage, processing, features, models

New A3 layer (blue)

Interval calibration, site and battery model, dispatch simulation

Honest status

Planned for A4: dashboard, scheduled runs and retraining

1,767,903

rows of 5 minute data in Master Meter 1

19

406 column names reconciled, yearly extracts profiled.

6

The register wraps repaired and validated against power with 99.7% accuracy.

30

hourly features, leakage safe

8

Quality gates with thresholds

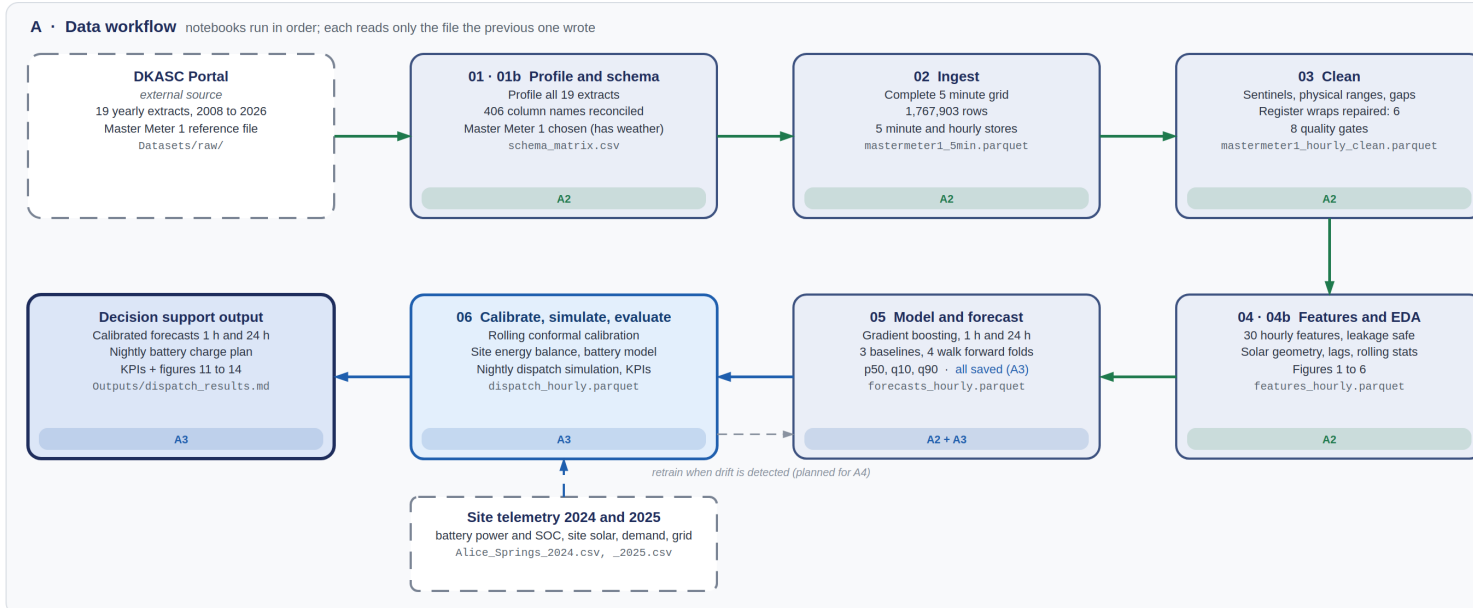
Parquet

columnar store at the 5 minute and hourly level

Workflow: data to decision, and how we work

PRT661 Solar Generation Forecasting · Workflow as Built at Assessment 3

How raw data becomes a battery decision, and how the team delivers each change

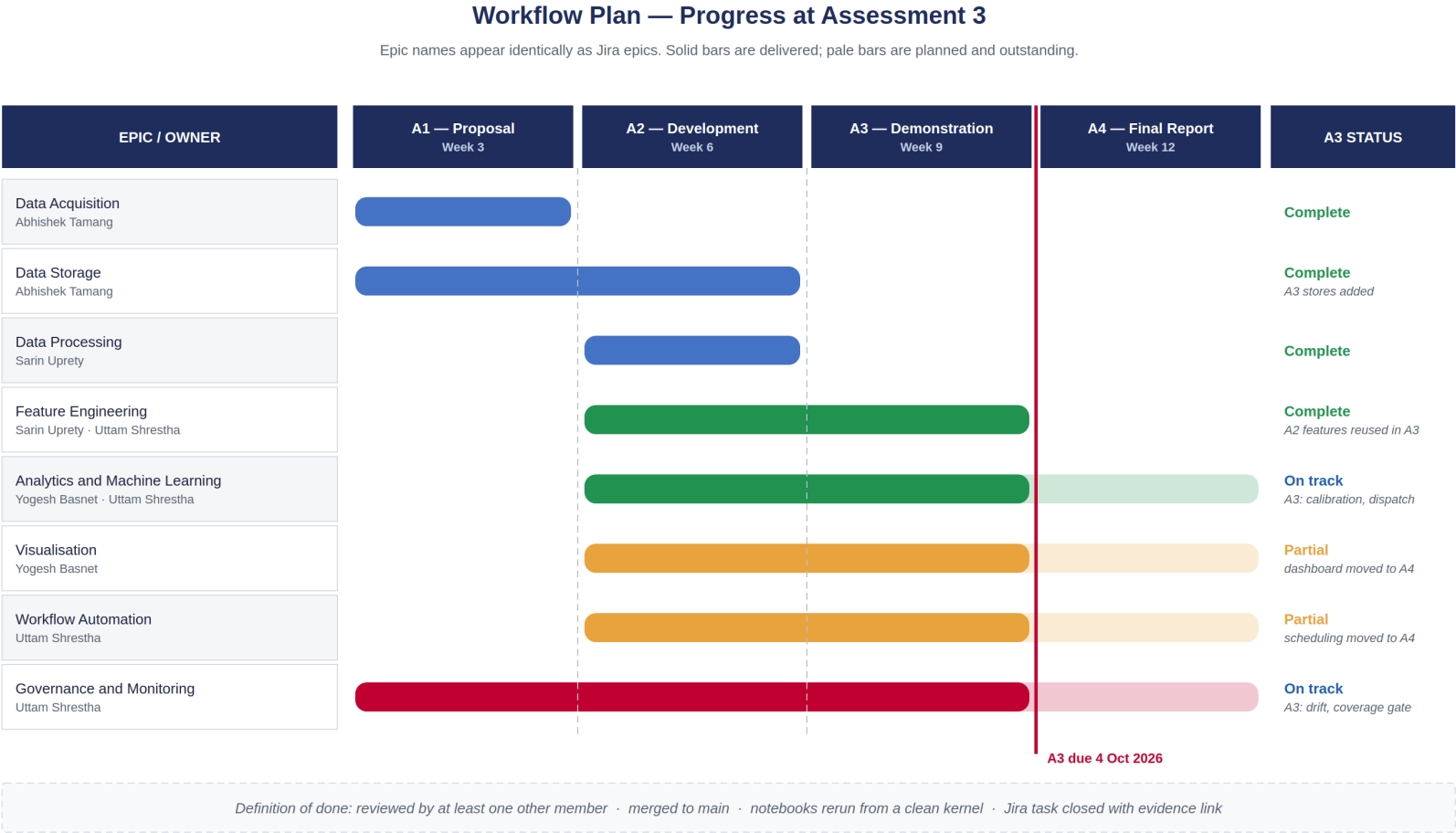


- A: notebooks 01 – 06 execute sequentially; only the previous notebook writes to its respective notebook.
- The blue boxes are new to A3: notebook 06 and the saved forecasts from 05
- B: All changes are done using the same 7 steps as for Jira task to merge
- Example: SCRUM-44 to 47, branch, pull request #2, merged

7

Implementation progress at A3

Sarin



16

A3 tasks in Jira Sprint 3

PR #2

A3 analysis merged into main

6 of 8

epics complete or on track

8 A2 recap: models and the open problem

Sarin

Model	1 h MAE	1 h R ²	24 h MAE	24 h R ²
Gradient boosting	3.61 kW	0.980	6.96 kW	0.913
Seasonal naive	7.74 kW	0.887	7.74 kW	0.887
Persistence	13.90 kW	0.866	7.74 kW	0.887
Climatology	11.07 kW	0.889	11.07 kW	0.889

Mean of four walk forward folds, 2019 to 2025

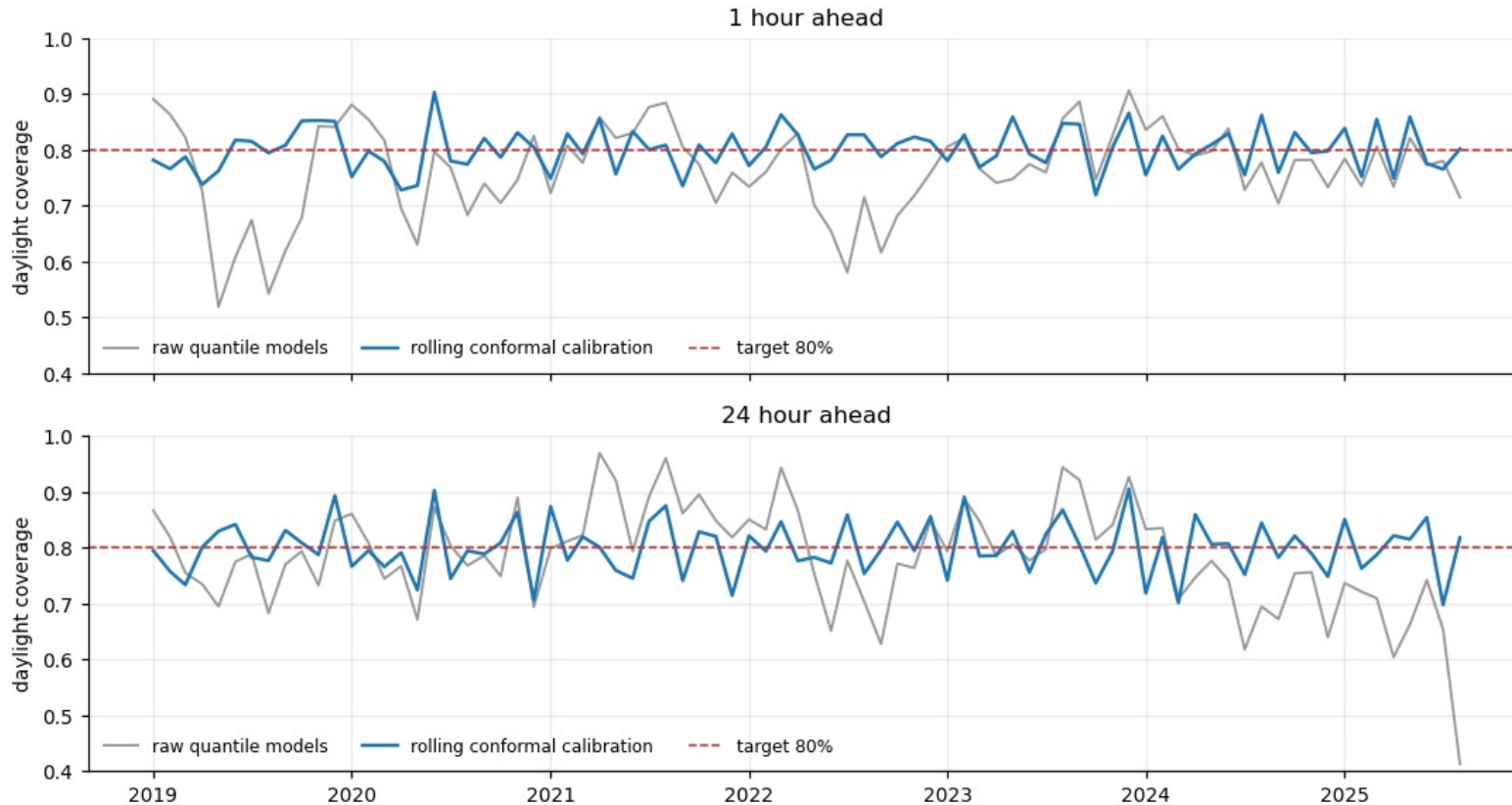
Carried into A3

70.6%

of daylight hours inside the 24 h 80% interval, latest test period 2024 to 2025 (target 80%)

The interval was too narrow: in 10 trials the interval was correct only 7 times but in 10 trials it had 80% confidence. A battery planning based on this would be overconfident.

A3 Part A: calibrating the forecast intervals



80.1% · 80.0%

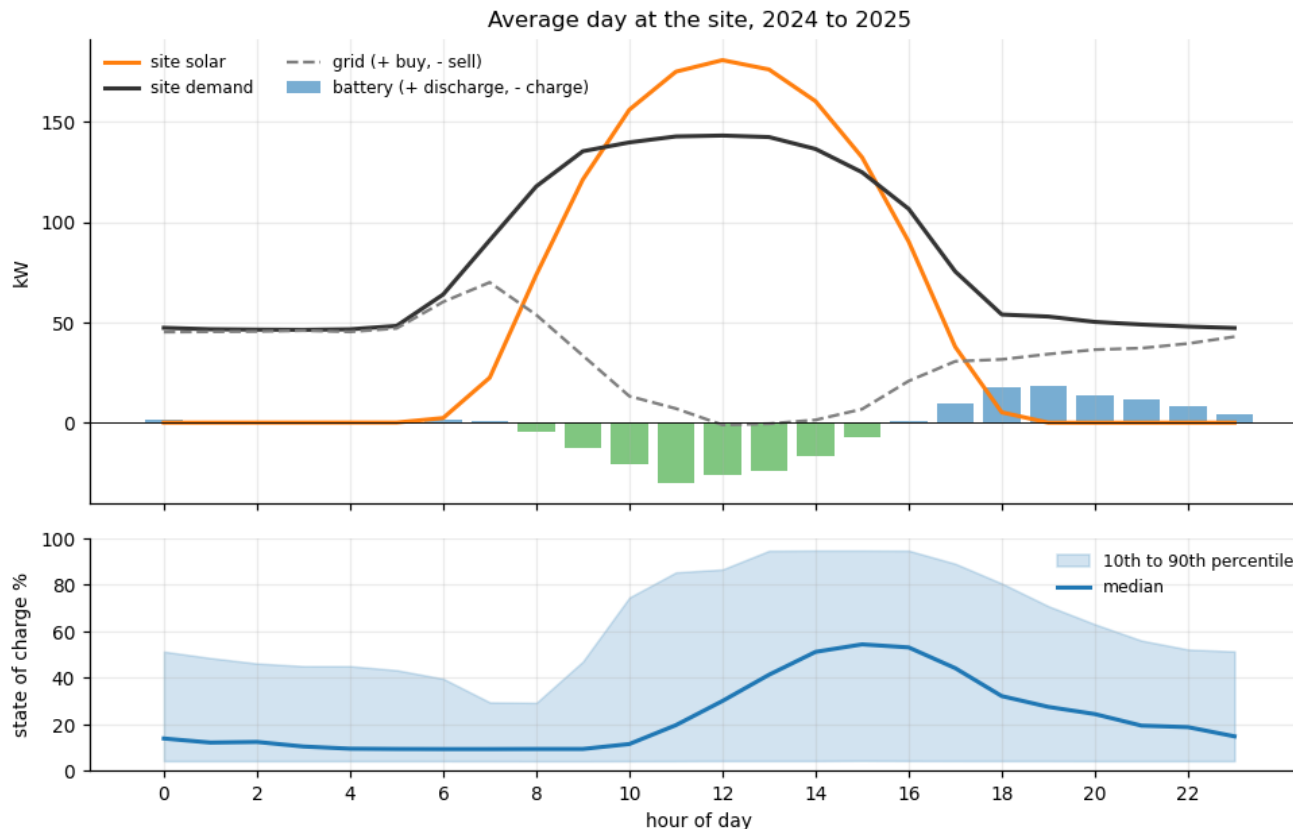
daylight coverage 1h and 24h for all four folds,
target 80%

- Fixed 3929 forecasts where the low edge was above the high edge
- Rolling conformal calibration: expand or contract over the past 30 days of errors
- Only uses data known at the moment, no look ahead
- Latest period, 24 h: 70.6% before, 79.3% after
- Every fold now 79.3% to 81.0%

Monthly daylight coverage, raw (grey) vs calibrated (blue)

Method: Rolling Adaptation for Conformalised Quantile Regression

A3 Part B: the site, its battery, and data drift



demand = solar + grid + battery

The energy balance test corrected the sign of all meters (96% of hours within 2 kW)

354 kWh

battery capacity via
telemetry

115 kW

charge and discharge
power

82.8%

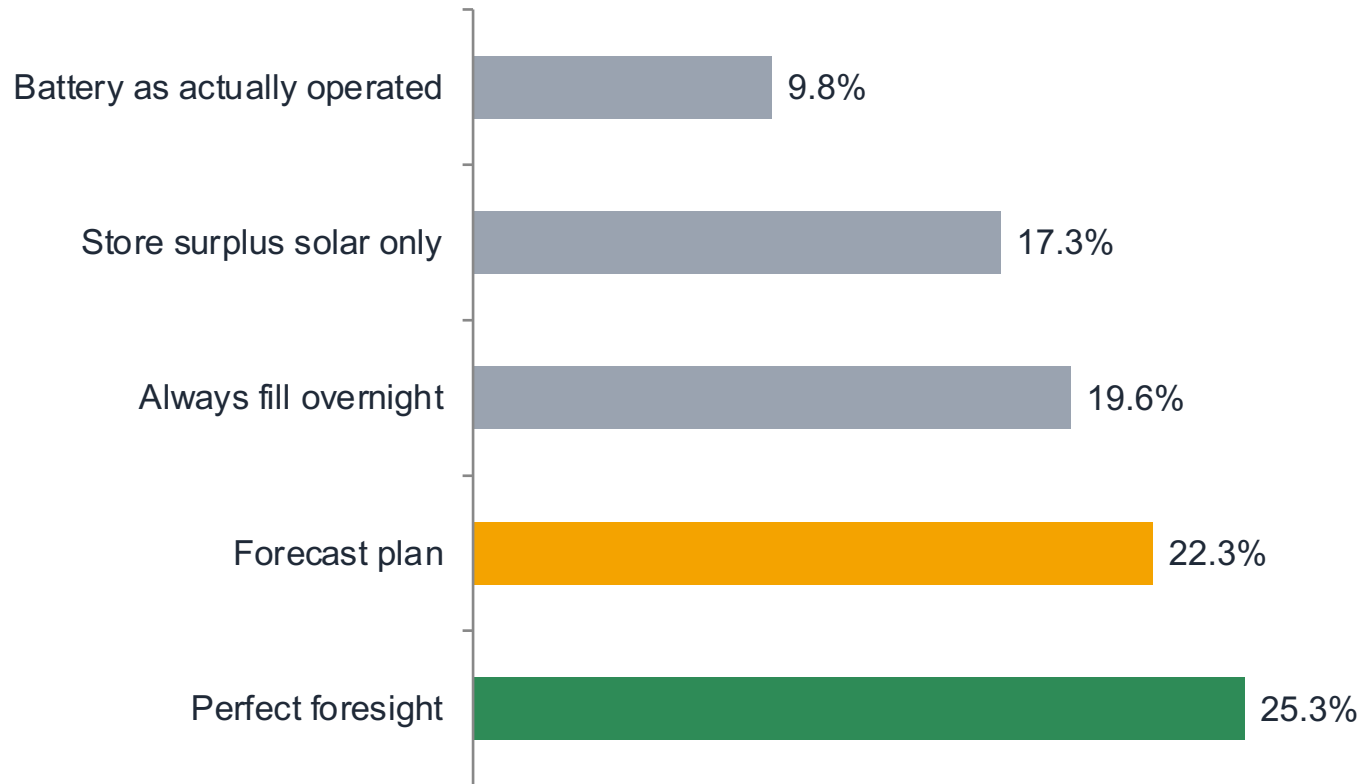
Round trip efficiency in
real cycles.

Data drift, end of February 2024

The site was reconfigured: site solar divided by Master Meter 1 dropped from 2.21 to approximately 1.1. Analysis begins from 1 March 2024 and the scale used is $K = 1.135$ (correlation 0.995 after that).

A3 Part C: the nightly battery decision

Apr 2024-Aug 2025, Saving against no battery.



At 22:00 every night

battery space minus tomorrow's forecast solar surplus = grid energy to buy tonight

\$3,419

saved beyond the best rule with no forecast

47%

of the benefit perfect foresight would give

Illustrative time of use tariff: \$0.15 off peak, \$0.25 shoulder, \$0.40 peak, \$0.07 feed in



Jupyter · notebook 06

calibration table, example week (Figure 14), final KPI table



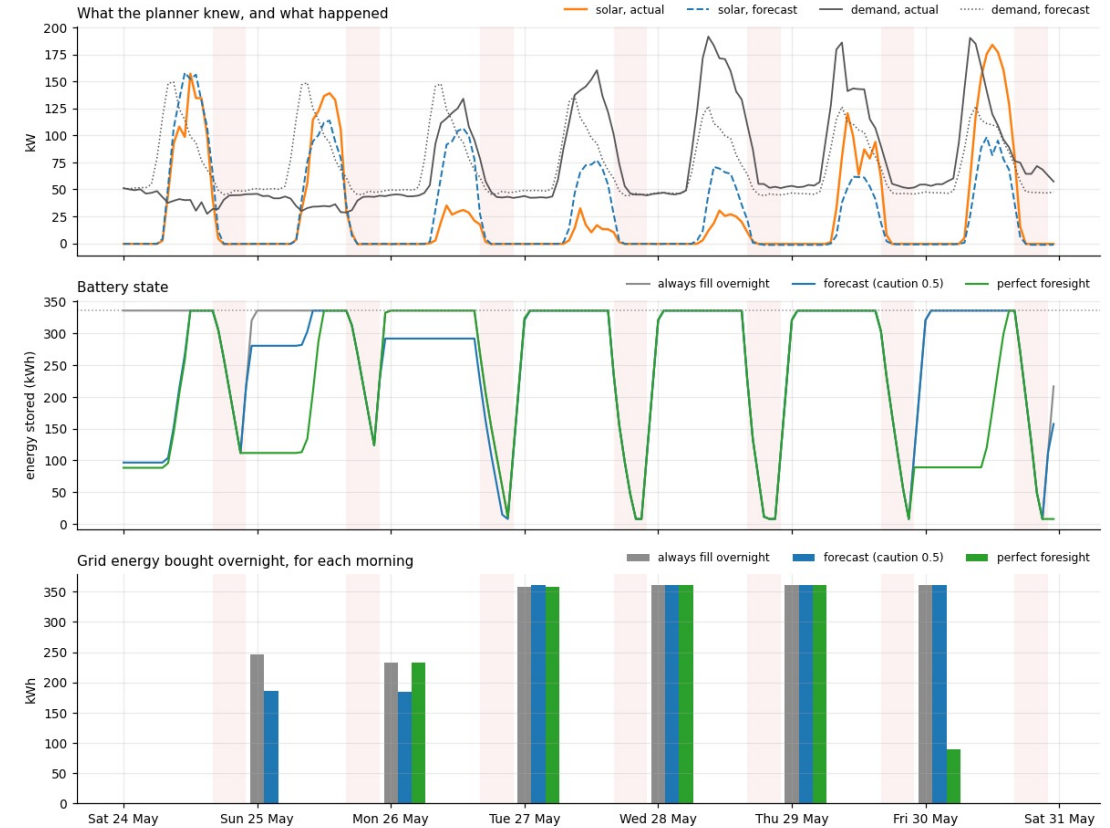
GitHub · pull request #2

commits with SCRUM keys, notebooks, figures, diagrams in the repo



Jira · Sprint 3

A3 tasks with owners and status





Crossed interval edges

Problem: Went low edge on high edge in 3,929 forecasts, primarily at night.

Fix: Re-order the two sides, the median remains the same.



Overconfident intervals

Problem: In the most recent time period, 24 h range covered 70.6% rather than 80%.

Fix: Rolling conformal calibration (there: 79.3%, overall: 80.0%)



Data drift

Problem: Solar ratio reduced by half from end of February 2024 due to site reconfiguration.

Fix: Start in March 2024, scale fitted on March only



Unknown signs, logger outages

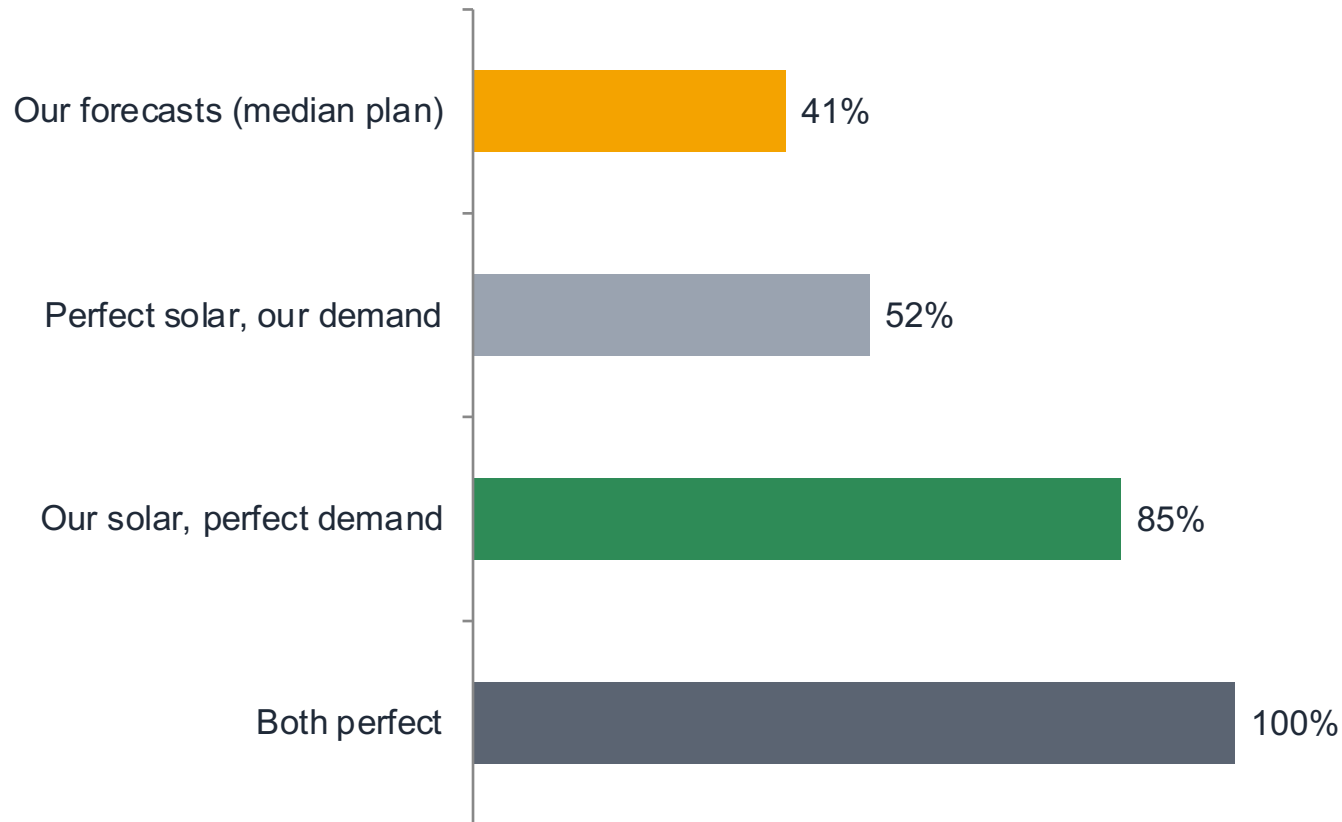
Problem: Meters are not specifying direction of positive, SOC read 0 in 7% of hours.

Fix: Energy balance test corrected for sign; excluded outage hours.

What is the limiting factor

Yogesh

What's holding the battery plan back?



Swaps by one piece at a time, based on perfect information, using the median plan (41%). The caution selected on 2024 is 47% identical to slide 11.

The weak link is "demand"

Perfect demand increases plan to 85%, perfect solar to 52%

Caution barely matters

Chosen on 2024, tested on 2025: 48% to 50% for caution 0 to 1

The larger the battery, the greater the value

Forecast gain over rules: 2.2 at half size, 3.9 at double

1

Better demand forecast

weekday and weekend patterns, temperature

2

Optimiser instead of a rule

plan charging and discharging together

3

Store more solar

113.6 MWh was cheaply exported rather than stored

4

Drift monitoring and retraining

Automatically detects changes, such as February 2024.

5

Final report and dashboard

Week 12: A4 professional report

Summary

80%

calibrated intervals that mean what they say

354 kWh

The battery is modelled using its own data.

22.3%

saving with a forecast planned battery

GitHub github.com/Uttam1618/PRT661-solar-forecasting

Jira uttamshrestha1618.atlassian.net, project SCRUM

Thank you